What I have written down is a diagram of a chemical reaction network. It indicates that a molecule of A can decompose into two molecules of B, that two molecules of B can react to form one molecule of A, that a molecule of A can react with a molecule of C to form a molecule of D, and so on.

Now suppose that I throw various amounts of my species into a pot. I am going to presume that the pot is stirred constantly so that its contents remain spatially homogeneous for all time, and I shall also suppose that the contents of the pot are forever maintained at fixed temperature and total volume. This, of course, is not to say that the chemical composition of the mixture within the pot will remain constant in time, for the occurrence of chemical reactions will serve to consume certain species and generate others. In fact it is the temporal evolution of the composition that we wish to investigate. With this in mind we denote the (instantaneous) values of the molar concentrations of the species by $c_A(t)$, $c_B(t)$, $c_C(t)$, $c_D(t)$ and $c_E(t)$, and we abbreviate this list of numbers by the "composition vector" $c(t)$.* Thus the picture we are thinking about, at least for the moment, looks something like that shown in Figure 1.1.†

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*A molar concentration, say c_A, specifies the number of A molecules per unit volume of mixture. More precisely, c_A is the number of A molecules per unit volume divided by Avogardro's number, 6.023×10^{23}. We shall be somewhat more precise about what we mean by the "composition vector" in Section 1.B.

†The reactor depicted in Figure 1.1 is closed with respect to the exchange of matter with its environment. Our focus on such reactors is temporary and is merely intended to illustrate in a simple context how chemists and engineers formulate differential equations based upon a set of reactions believed to approximate the true chemistry. We will begin to consider "open" reactors in the next lecture (Section 2.E). There we shall indicate how open reactors can be modelled in terms of reaction networks and how the appropriate differential equations, like those for closed reactors, bear a definite relationship to reaction network structure.